

Bank Management System

Individual Requirements

ID	Requirement	Category
FR-1	The system shall allow a customer to open a new account by submitting KYC details (name, address, phone, ID proof) and an initial deposit of at least ₹500.	Functional
FR-2	The system shall generate a unique 10-digit account number for every new account.	Functional
FR-3	The system shall allow staff/admin users to view, edit, or deactivate an existing customer account.	Functional
FR-4	The system shall verify that a withdrawal amount does not exceed the available balance minus a ₹1,000 minimum balance, and reject the transaction with an "Insufficient balance" message if it does.	Functional
FR-5	The system shall generate a unique transaction reference ID and receipt for every completed deposit or withdrawal.	Functional
FR-6	The system shall lock a user account for 15 minutes after 3 consecutive failed login attempts.	Functional
FR-7	The system shall log every deposit and withdrawal with a timestamp, transaction type, amount, and post-transaction balance.	Functional
FR-8	The system shall allow an admin to generate a report of total deposits, withdrawals, and active account count for a given date range within 10 seconds.	Functional
NFR-1	The system shall complete 95% of deposit and withdrawal transactions within 2 seconds, including under a load of up to 500 concurrent users.	Non-Functional (Performance)
NFR-2	The system shall support at least 1,000 concurrent authenticated sessions without response time exceeding 4 seconds.	Non-Functional (Scalability)
NFR-3	The system shall encrypt all stored passwords and PINs using a one-way hashing algorithm (e.g. bcrypt).	Non-Functional (Security)
NFR-4	The system shall reject any transaction that would result in a negative account balance or a duplicate account number.	Non-Functional (Data Integrity)